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Preliminary survey of pests and diseases of eucheumatoid seaweed farms in the Philippines

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Abstract

Farmed eucheumatoids are the top aquaculture commodity in the Philippines, contributing more than 60% of total national aquaculture production by volume. Despite significant production losses observed in recent years due to pests and diseases, data relating to the temporal and spatial geographic distribution, seasonality, prevalence, and etiology of these critical problems at the farm level are lacking. Here, we assessed the prevalence of pests and diseases at 16 farm sites within several major seaweed growing areas, including Luzon (Palawan), Visayas (Bohol), and Mindanao (Zamboanga City, Davao del Norte, and Tawi-Tawi). Prevalence of pests and diseases were recorded by assessment of individual seaweed plants in situ, comprising ≥5% of the total number of seaweed plants on each farm. Results from this survey revealed "ice-ice" disease (IID) at all sites, albeit at low prevalence rates of below 25%, the exception being one site where prevalence was 100%. No significant difference in the prevalence of IID was found between farms in shallow (≈0.5 m during low tide) and deep water sites (depths exceeding 4 m during the lowest tide) ($p > 0.05$). Moderate prevalence (26–50%) of the macro-epiphyte pests *Sargassum* and *Ulva* was reported in Luzon. Further, epiphytic filamentous algae (EFA), black spots, and evidence of grazing were recorded in several farms with low prevalence (0.2–11.2%). The three farms with the highest prevalence of IID and EFA were evaluated to determine the severity of cover on the thallus of infected plants. A very high severity of IID (76–100% coverage) was observed on the secondary and primary branches, while for EFA, low severity (≤25%) was found in all regions of the thallus. Our results suggest the widespread occurrence of yield-limiting diseases and pests affecting eucheumatoid farms in the Philippines. Strategies for the diagnosis, surveillance, and management of yield-limiting pests and diseases are needed to ensure maintenance and sustainable expansion of this important aquaculture sector in the Philippines.

Keywords Aquaculture · Eucheumatoids · Seaweed farming · Diseases · Pests

Introduction

Among the 6500 species of red algae (Rhodophyta) described worldwide (Guiry and Guiry 2020), only a few genera (e.g., *Gracilaria*, *Eucheuma*, and *Kappaphycus*) have been commercially exploited and successfully introduced to tropical and sub-tropical regions (Ask and Azanza 2002; Ask et al. 2003). Of these, *Kappaphycus alvarezii* (Doty) Doty ex.

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P.C.Silva, *Kappaphycus striatus* (Schmitz) Doty ex P.C.Silva, and *Eucheuma denticulatum* (Burman) Collins and Hervey are now cultivated in over 20 countries in Southeast Asia, the South Pacific, Latin America, and the Indian Ocean (Sulu et al. 2004; Pickering 2006; Hurtado et al. 2014; Msuya et al. 2014; Hayashi et al. 2017; Shanmugam et al. 2017; Alemañ et al. 2019). The exponential increase in production of these seaweeds over the past five decades is due to an increased demand for carrageenan, an extract valued for its hydrocolloid polysaccharides in the food, cosmetics, pharmaceutical, and nutraceutical industries (Bixler and Porse 2011; Porse and Rudolph 2017). In 2018, farmed eucheumatoid production reached 11 million tonnes (fresh weight), 34% of the total world production of aquatic plants (FAO 2020).

Seaweeds are an important aquaculture crop in the Philippines with production occurring in 15 of the 17 geographical regions of the country (Bureau of Fisheries and Aquatic Resources 2018). In 2018, seaweeds comprised 64% (equivalent to 1.48 Mt, fresh weight) of the total aquaculture production in the Philippines, making it the top aquaculture commodity by volume, and the second by value, supporting an export industry for carrageenan worth US$ 207 million (BFAR 2018). Seaweed farming has also become a primary source of income and employment in several remote areas of the country where livelihood opportunities, other than those provided by fishing, are limited (Hurtado 2013; Valderrama et al. 2013); currently, more than 200,000 families are engaged in seaweed farming in the Philippines, mostly in rural coastal areas, providing significant societal and economic value to these communities (Pedrosa 2017).

Despite its historical position as the top producer globally, the past decade has seen a decline in production of eucheumatoid seaweeds in the Philippines (Philippine Statistics Authority 2015, 2019). Factors such as unfavorable weather conditions and political unrest in farming areas have resulted in a decline in production; in particular, typhoons have damaged seaweed farms several times in the last three decades, and changing weather patterns have prevented production throughout the year (Hurtado 2013). At the same time, outbreaks of pests and diseases such as "ice-ice" disease (IID) and epiphytic filamentous algae (EFA) infestation have exacerbated production problems (Hurtado and Critchley 2006; Hurtado et al. 2006a; Vairappan et al. 2008).

Seaweeds affected by IID display a gradual loss of pigmentation followed by softening of the thallus, and detachment of diseased plants from the cultivation lines used to anchor the seaweeds at farm sites (Doty and Alvarez 1975; Largo et al. 1995; Arasamuthu and Edward 2018). EFA are red filamentous algae that attach to the surface of the host seaweeds, penetrating from the cortical to the medullary layers, causing damage to the host plant (Ask 1999; Vairappan 2006; Pang et al. 2011; Tsiresy et al. 2016). Both IID and EFA outbreaks have been reported to affect biomass production and the quality of carrageenan extracted from affected seaweed plants (Largo et al. 1995; Mendoza et al. 2002; Hurtado and Critchley 2006; Hurtado et al. 2006a, 2019; Vairappan et al. 2008; Ward et al. 2020). Such outbreaks and other factors resulted in mean production losses of 16.8% per annum between 2012 and 2018 compared to the peak of production of 1.84 Mt in 2011 (PSA 2013, 2015, 2019). However, in lieu of official surveillance, data on the occurrence of IID and other pests of seaweeds at the farm level in Philippines is wanting. A better understanding of pests and disease is required in order to implement an effective biosecurity strategy to protect this valuable commodity (Campbell et al. 2020). To address this, we designed a sampling strategy to document signs of pests and diseases in several major eucheumatoid farming areas in the Philippines and quantified the occurrence of IID, EFA, and other observations, including macro-epiphytes, black spots (black or brown raised lesions, c. 0.5 mm diameter, on the surface of the eucheumatoid thallus), and the evidence of grazing. We also evaluated the severity of IID and EFA at selected sites and report on significant potential for yield limitation due to these conditions within country.

Materials and methods

Sampling sites

Sixteen farming sites in different major seaweed producing areas of the Philippines were assessed (Table 1). The three geographical island groups of the country were represented: Luzon—Palawan (n=5); Visayas—Bohol (n=5); and Mindanao—Zamboanga City, Davao del Norte, and Tawi-Tawi (n=6) (Fig. 1). Sampling activities were conducted within the period October 2018 to October 2019 and coincided with the southwest monsoon season.

Sampling design and assessment

Seaweed farms were chosen based on the biomass output in the area (BFAR 2017), accessibility, and availability of assistance and a local person as a guide during assessment activities. The following information was gathered from the farm owners/operators: the culture technique, species farmed, the number of cultivating lines per farm. Based on this information, four to five cultivating lines to which individual seaweeds were attached were randomly selected for assessment, representing at least 5% of the total individual seaweeds at each farm site. In situ visual inspection for the presence of diseases, pests, and other abnormalities, including black spots and grazing, was made on plants from each line. Assessments were conducted during low tides for fixed off-bottom (shallow water) farms, and a motorized boat was used to access farms in

Table 1 Sampling site locations in major seaweed farming areas in the Philippines, to include farming technique, seaweeds density, and species of seaweeds farmed (n=16)

Sampling sites	Grid reference	Farm location	Farming technique	Density seaweeds		Species farmed		
				Per line	Per meter	*Kappaphycus alvarezii*	*Kappaphycus striatus*	*Eucheuma denticulatum*
Luzon								
Palawan								
Puerto Princesa	9.97305° N, 118.944231° E	Deep water	Triangular	3000	10	+	-	-
Balabac	8.303103° N, 117.182876° E	Shallow water	Fixed off-bottom	92	4	+	+	-
Quezon	9.347657° N, 118.115874° E	Deep water	Hanging long-line	425	4	+	-	-
Pamantolon, Taytay	10.89762° N, 119.499552° E	Deep water	Hanging long-line	495	4	+	+	+
Pularaquen, Taytay	10.959025° N, 119.510933° E	Deep water	Hanging long-line	300	4	+	+	-
Visayas								
Bohol								
Danahon Reef	10.278365° N, 124.5682° E	Shallow water	Fixed off-bottom	61	4	-	-	+
Mahanay	10.231802° N, 124.260072° E	Shallow water	Fixed off-bottom	146	4	-	+	+
Guindacpan	10.224459° N, 124.286356° E	Deep water	Hanging long-line	224	4	-	+	+
Jao 1	10.173747° N, 124.385620° E	Shallow water	Fixed off-bottom	155	4	-	+	-
Jao 2	10.173747° N, 124.385619° E	Shallow water	Fixed off-bottom	202	4	-	+	-
Mindanao								
Zamboanga City								
Tigtabon	6.898328° N, 122.173129° E	Deep water	Multiple raft, long-line	49	4	+	+	+
Layag-Layag	6.886975° N, 122.117179° E	Deep water	Multiple raft, long-line	124	4	-	+	-
Arena Blanco	6.913056° N, 122.163333° E	Deep water	Multiple raft, long-line	166	4	+	+	-
Davao del Norte								
Samal	6.969799° N, 125.721994° E	Deep water	Multiple raft, long-line	108	4	+	-	-
Tawi-Tawi								
Simunul	4.919157° N, 119.825235° E	Shallow water	Fixed off-bottom	97	4	-	+	-
Tundon	4.912309° N, 119.944064° E	Shallow water	Fixed off-bottom	73	4	-	+	-

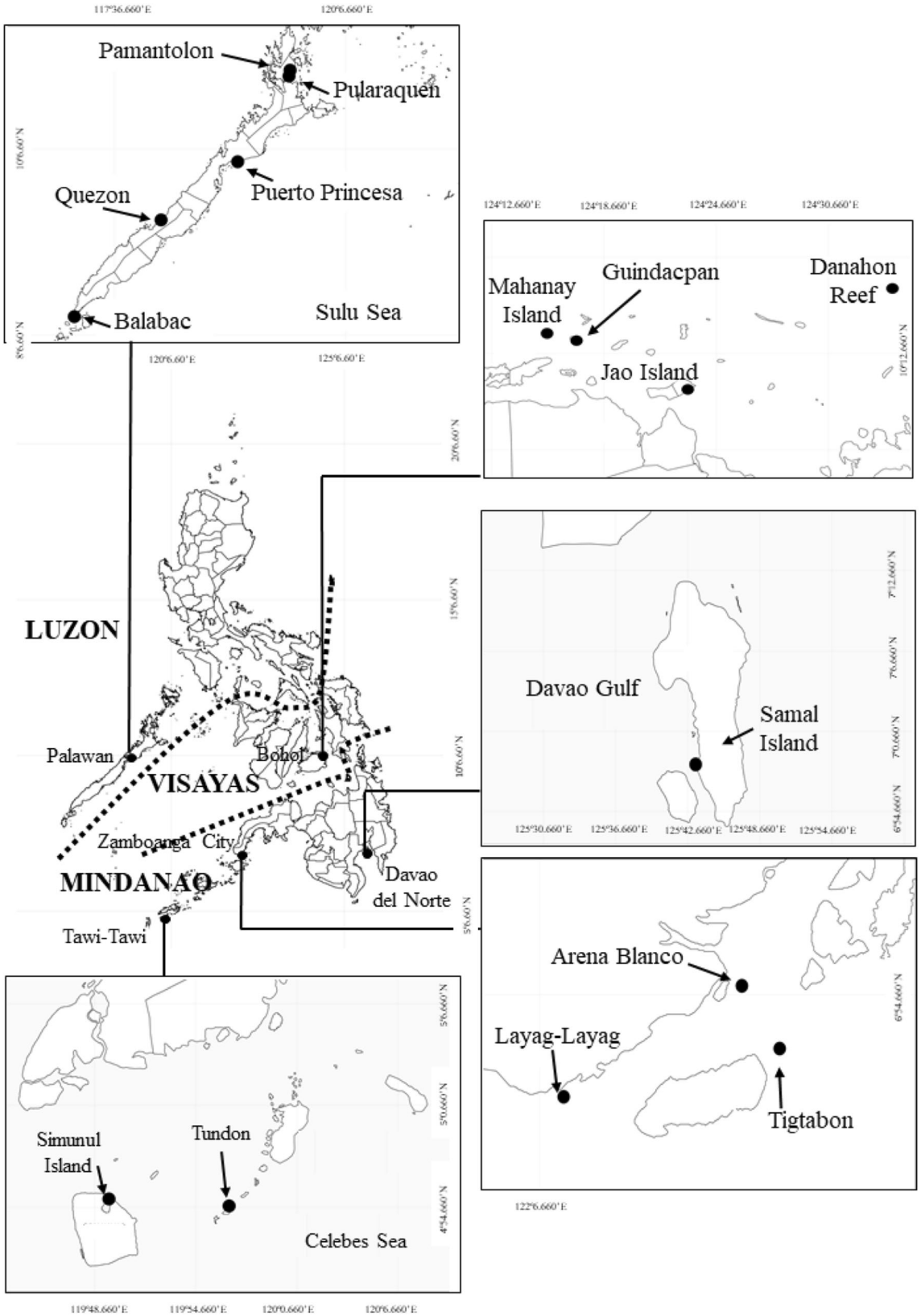


Fig. 1 The Philippine map showing the sampling site locations in major seaweed growing areas in the Philippines. Luzon (Palawan; *n*=5), Visayas (Bohol; *n*=5), and Mindanao (Zamboanga City, Davao del Norte, and Tawi-Tawi; *n*=6)

deeper waters. The prevalence of pests, diseases, and other abnormalities was calculated:

$$\text{Prevalence } (\%) = \frac{\text{number of affected seaweeds}}{\text{total number of seaweeds assessed}} \times 100$$

Prevalence was differentiated into low (≤25%), moderate (26–50%), high (51–75%), and very high (76–100%).

In addition, the severity cover (%) measured the degree of infection or infestation on the different parts of the thallus. Seaweed samples from the three farms with the highest prevalence of IID (n=120), and EFA (n=84) were used in this assessment. For each sample, the apical, secondary branch, and the primary branch of an individual seaweed thallus were assessed (Fig. 2a). The presence of IID and EFA was estimated as percentage cover using a semi-quantitative scale of 1–5 (1 = 1–10%, 2 = 11–25%, 3 = 26–50%, 4 = 51–75%, and 5 = 76–100%). The total score per region of tissue was divided by the number of seaweeds assessed to obtain a per-plant severity score: classified into four infection/infestation covers of low (≤25%), moderate (26–50%), high (51–75%), and very high (76–100%).

Pest identification

Pests were categorized initially as macro-epiphyte alga, i.e., seaweeds that grow on the surface of seaweed plants, or as epiphytic filamentous algae (EFA), i.e., seaweeds which grow from the cortical layer (Ask 1999). Epiphytes were further classified based on the mechanism of attachment (weak or strong), and penetration (outer or inner), as described by Leonardi et al. (2006) and Ingle et al. (2018). Macro-epiphytes present in farms were identified through morphological characteristics and confirmed using reference manuals (Hurtado et al. 1992, 2006b). To determine the degree of epi-

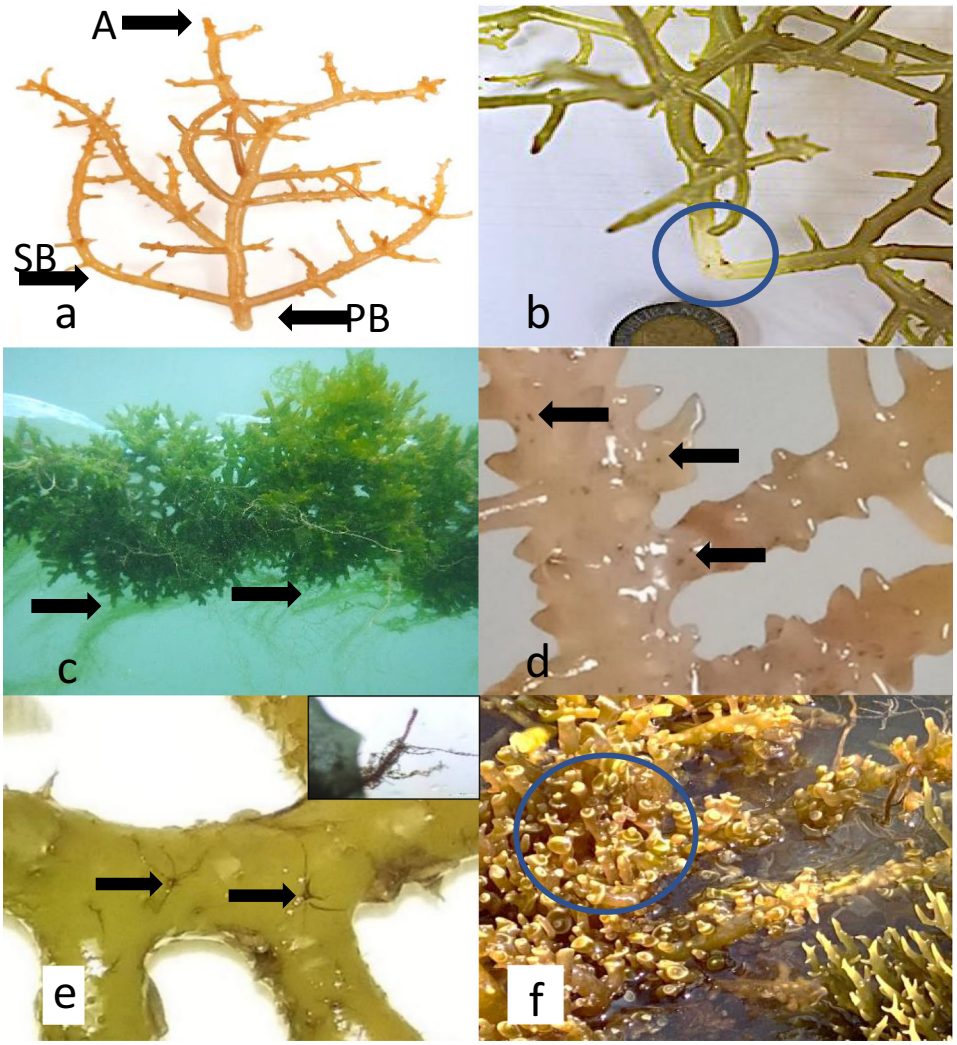


Fig. 2 Observations on the farmed seaweeds: (a) showing the different tissue regions of the seaweeds assessed, including the apex (A), the secondary (SB), and primary branches (PB) (arrows), (b) *K. striatus* affected by "ice-ice" disease (IID) in the primary branch (encircled), (c) filamentous green *Ulva* entangled loosely on *K. striatus* (arrows), (d) *E. denticulatum* thallus affected by black spots (arrows), (e) epiphytic filamentous algae (EFA) (arrows) affecting the thallus of *K. striatus*, inset showing the penetration of the epiphytes to the cortical and medullary tissues, and (f) grazing on *K. alvarezii* manifested by the absence of tips on the apex soft tissues (encircled) (Photos courtesy of AQ Hurtado and RC Sibonga)

phyte penetration to the host plant, fresh transverse-sections (approx. 0.50 mm thick) of seaweeds affected by EFA were removed using a surgical blade and placed on a glass slide before viewing under the light microscope (Olympus CX43, Japan) connected to a camera (Olympus DP2-SAL, Japan).

The observations of diseases and pests in this study were categorized according to the nature of the association with the host seaweed: (1) IID and black spots (result in changes to thallus tissue), (2) macro-epiphytes (non-penetrating, but attached or entangled), (3) epiphytic filamentous algae, EFA (penetrating the thallus), and (4) grazing (mechanical damage or removal of tissues).

Environmental parameters

Water parameters analyzed in situ were dissolved oxygen and temperature (EcoSense DO200A, YSI, USA), irradiance (LI-250A Photometer LI-COR Biosciences, USA), depth (Hawkeye H22PX Handheld Sonar System, USA), salinity (Digital refractometer MA886 Sodium Chloride Refractometer, Milwaukee Instruments, Inc., USA) and pH (pH meter PH-200 HM Digital, USA). The collection of water parameters was conducted mainly in the morning except in Pularaquen, Taytay, and Simunul, Tawi-Tawi, when it was conducted in the afternoon. Additional water samples were collected from each farm to be analyzed for nutrient levels, including nitrite, nitrate, ammonia and phosphate (LaMotte test kits, USA). The samples were transported and stored on cool packs (4°C) in the dark (i.e., in an icebox) throughout the sampling period until analysis. Monthly climatological data, including amount of total rainfall, mean wind speed, and mean near-surface temperature (T_{mean}), were collected from the nearest synoptic station provided by the Department of Science and Technology Philippine Atmospheric, Geophysical and Astronomical Services Administration (DOST PAGASA) except for Tawi-Tawi farms with no near synoptic station. In Luzon (Palawan) farms, the nearest synoptic stations are located in Puerto Princesa and Coron. In the Visayas (Bohol) farms, all the climatological data were collected from Mactan Station. While in Mindanao farms, the closest synoptic stations are located in Davao City and Zamboanga City.

Statistical analysis

Farms were grouped depending on their geographical island group location: Luzon (Palawan; n=5), Visayas (Bohol; n=5), and Mindanao (Zamboanga City, Davao del Norte, and Tawi-Tawi; n=6). Mean prevalence per farm and geographic area were expressed as percent mean ±SEM. After prevalence rates were found not to be normally distributed using Shapiro-Wilk's test and the variances were not homogeneous, non-parametric analysis of variance (ANOVA) Kruskal-Wallis test was used to check significant differences among farms within each island group as well as across the three island groups. A non-parametric Mann-Whitney U test was also used to compare the difference between the location of farm sites (shallow vs deep water) in relation to IID prevalence. A level of significance $p \leq 0.05$ was used as the threshold for significant differences. All statistical tests were undertaken using SPSS ver. 23 (SPSS Inc., USA). No statistical analyses were conducted on the environmental water parameters since the sampling was done only once during the assessment activity.

Results

Farming sites and seaweed cultivars

Seaweed farms were found either in shallow or deep water areas. Both the shallow and deeper water farms were represented in each region. Four culture techniques were employed: triangular (1 farm), fixed off-bottom (7 farms), hanging long-line (4 farms), and multiple raft long-line (4 farms). Shallow water sites were accessed from the shoreline in a minimum water depth of 0.5 m during the lowest tide, while deeper sites were generally in depths in excess of 4 m during the lowest tide. The number of seaweeds varied from each farm and ranged from 49 to 3000 seaweeds per line depending on the length of the cultured line and the culture techniques employed. Cultivation density (number of seaweeds per meter of line) was 4 seaweeds per meter for all farm sites studied, with the exception of one farm in Palawan, where the cultivation density was 10 seaweeds per meter. *Kappaphycus alvarezii*, *K. striatus*, and *E. denticulatum* were the common seaweed species being cultured (Table 1).

Diseases and pests occurrence

"Ice-ice" disease (IID) and pests (macro-epiphytes and EFA) were commonly observed, as were other abnormalities such as black spots and grazing wounds (Table 2). The descriptions of the pest and disease observations are reported in Table 3.

"Ice-ice" disease (IID)

Seaweeds displaying IID were found to have lost pigmentation in patches of the thallus. Depigmented regions appeared light pink to white in color, notably at locations in the middle and base of the thalli (Fig. 2b). Depigmentation appeared to be a progressive condition, spreading to other regions of the thallus in many of the affected specimens observed. In more severe cases, softened, necrotic, and decomposing areas of tissue were observed leading to loss of attachment of seaweeds from the cultivation lines. In Luzon, the Quezon farm site showed

Table 2 Summary of diseases and pest prevalence (%) in farms across different geographic island groups: Luzon (n=5), Visayas (n=5), and Mindanao (n=6). Sampling sites with different superscripts within geographic group denotes significant difference at p<0.05. Means are expressed as ±SE

Sampling sites	"Ice-ice" disease (IID)	Epiphytic filamentous algae (EFA)	Macro-epiphytes			Black spots	Grazing
			Sargassum	*Ulva*	*Gracilaria*		
Luzon							
Palawan							
Puerto Princesa	2.4±0.2^{ab}	0^{b}	0.25.1±0.3^{ab}	0.2±0.3^{ab}	0	0	0
Balabac	1.3±1.3^{ab}	0^{b}	0^{b}	0^{b}	0	0	0
Quezon	8.5±2.0^{a}	0.2±0.0^{a}	0.1±0.1^{ab}	0^{b}	0	0	0
Pamantolon, Taytay	0.3±0.0^{ab}	0^{b}	33.0±7.9^{a}	17.1±5.6^{ab}	0	0	0
Pularaquen, Taytay	0.2±0.1^{bc}	0^{b}	41.3±9.5^{a}	44.3±10.2^{a}	0	0	0
Visayas							
Bohol							
Danahon Reef	3.5±1.6^{ab}	0^{b}	0	0	0	0^{b}	0^{b}
Mahanay	8.4±2.2^{a}	0^{b}	0	0	0	0^{b}	4.6±1.5^{a}
Guindacpan	0.7±0.3^{b}	4.5±1.7^{a}	0	0	0	1.7±0.6^{a}	0^{b}
Jao 1	2.2±0.7^{ab}	0^{b}	0	0	0	0^{b}	0^{b}
Jao 2	1.7±1.1^{ab}	0^{b}	0	0	0	0^{b}	0^{b}
Mindanao							
Zamboanga City							
Tigtabon	0.4±0.4^{b}	3.7±0.4^{ab}	0	8.6±0.4^{a}	0.8±0.5^{a}	0	0^{b}
Layag-Layag	0.6±0.2^{b}	0^{b}	0	0^{b}	0^{a}	0	0^{b}
Arena Blanco	10.2±0.0^{ab}	11.2±0.4^{a}	0	0.8±0.2^{ab}	0^{a}	0	0^{b}
Davao del Norte							
Samal	100±0.0^{a}	0^{b}	0	0^{b}	0^{a}	0	0^{b}
Tawi-Tawi							
Simunul	14.9±4.4^{ab}	0^{b}	0	0^{b}	0^{a}	0	4.8±2.4^{a}
Tundon	2.5±2.5^{b}	0^{b}	0	0^{b}	0^{b}	0	0^{b}

a significantly higher prevalence of IID compared with the farm in Pularaquen, Taytay (p=0.025), but did not differ significantly with the other farms in the area. In the Visayas, Mahanay recorded a significantly higher IID (p=0.015) compared with Guindacpan but not with the farms in Jao 1, Jao 2, and Danahon Reef (p>0.05). In Mindanao, the Samal farm recorded a very high IID prevalence of 100±0.0%, significantly higher than at farms in Tigtabon, Layag-Layag, and Tundon (p<0.05), but did not differ significantly to farms in Arena Blanco, and Simunul. Overall, prevalence rate was classified as low (≤25%) at all farm sites with the exception of the farm in Samal where prevalence was classified as very high (100%).

Macro-epiphytes

Macro-epiphytes were often found entangled but loosely attached to farmed seaweeds. The presence of macro-epiphytes, including *Sargassum* (brown alga), *Ulva* (green alga), and *Gracilaria* (red alga), was observed in Luzon and Mindanao farms; however, no macro-epiphytes were recorded on seaweeds observed at the Visayas sites (Table 2). *Sargassum* was present only in Luzon at four out of the five farms. Prevalence rates were classified as moderate in Pamantolon and Pularaquen, Taytay (26–50%). However, the results were not significantly different from the observations in farms in Puerto Princesa and Quezon. Filamentous *Ulva* species were observed in Luzon and Mindanao farms at three out of five, and two out of six farms, respectively. In Luzon, the farm in Pularaquen, Taytay recorded a moderate prevalence rate (44.3±10.2%), but the prevalence did not differ significantly from the farms in Pamantolon, Taytay and Puerto Princesa with low prevalence rates (≤25%). In Mindanao, *Ulva* was observed attached to seaweeds in Tigtabon and Arena Blanco farms (both in Zamboanga City) with low prevalence rates (Fig. 2c). *Gracilaria* was observed only in one farm in Mindanao with a low prevalence rate.

Table 3 Descriptions of important pests and diseases occurring in eucheumatoid crops in seaweed farms in the Philippines

Association with the host seaweed	Pest or disease	Description	References
Type 1	"Ice-ice" disease (IID)	Loss of pigmentation followed by softening of the thallus, and detachment of diseased plants from seaweed cultivation lines	Doty and Alvarez (1975); Largo et al. (1995); Arasamuthu and Edward (2018)
Type 1	Black spots	Black or brown raised spots, c. 0.5 mm diameter, on the surface of the eucheumatoid thallus	This paper
Type 2	Macro-epiphyte algae	Seaweeds that grow on the surface of seaweed plants (loosely attached or entangled), e.g., *Sargassum, Ulva, Gracilaria*	Leonardi et al. (2006); Ingle et al. (2018)
Type 3	Epiphytic filamentous algae (EFA)	Red seaweeds that grow on the surface of seaweed plants, penetrating from the cortical to the medullary layers, causing mechanical damage to the host plant, e.g., *Melanothamnus* (= previously known as *Neosiphonia*), *Polysiphonia*	Ask (1999); Vairappan (2006); Pang et al. (2011); Tsiresy et al. (2016)
Type 4	Grazing	Mechanical damage or absence of tips on the soft tissues at the apex of the thallus	Ask (1999); Pang et al. (2015)

Epiphytic filamentous algae (EFA)

Red epiphytic filamentous algae (EFA) were found attached by holdfasts to host plants in several farms. Affected seaweeds showed epiphytes penetrating the cortex and growing within medullary tissues (Fig. 2e). EFA were observed in four of the 16 farms assessed, including Quezon (Luzon), Guindacpan (Visayas), and Tigtabon and Arena Blanco (Mindanao). All the farms exhibited a low prevalence of EFA (≤25%).

Black spots

Black spots were defined as dark brown-black, irregular-shaped lesions, generally of <1 mm in diameter, on the surface of the seaweed thallus (Fig. 2d). They were only observed on plants collected from one farm in the Visayas, with a low prevalence rate (1.7±0.6%).

Grazing

Signs of grazing by marine organisms were observed principally on the soft tissues at the apex of the thallus. Signs of grazing were often manifested by the absence of tips (Fig. 2f). Signs of grazing were recorded in two shallow water farms, one in the Visayas and the other one in Mindanao, though the prevalence rate at farms in both areas was low (≤25%).

Summary of prevalence rates

When results from all farms were grouped according to geographical locations, the prevalence rates for pests, diseases, and other abnormalities were generally categorized as low (≤25%) (Fig. 3). The results for IID showed that Mindanao had a higher (21.4±6.6%) prevalence compared with Luzon (2.5±0.8%) and Visayas (3.3±0.7%), but no significant difference among the regions. Also, no significant difference in IID prevalence was observed when shallow water (5.0±1.1%) and deeper water (14.8±5.1%) farm sites were compared ($p>0.05$) (Fig. 4). The occurrence of pests, including macro-epiphytes and EFA, was observed in Luzon, Visayas, and Mindanao regions. For the macro-epiphytes, *Sargassum* was observed in Luzon at 14.9±4.8% but absent in Visayas and Mindanao farms. *Ulva* was present in Luzon (12.3±4.5%) and Mindanao farms (1.6±0.6%) only, and *Gracilaria* was recorded in one farm in Mindanao at 0.1±0.1%. For EFA, Mindanao recorded the highest prevalence rate at 2.5±0.8%, followed by Visayas and Luzon with 0.9±0.5%, and 0.1±0.0%, respectively. Black spots were only recorded in Visayas at a prevalence of 0.3 ±0.2%. Signs of grazing were recorded in Visayas (0.9 ±0.5%) and Mindanao (0.8±0.5%).

Severity of cover of IID and EFA

The three farms with the highest prevalence of IID were located in Mindanao, including Samal (100±0.0%), Simunul (14.9±4.4%) and Arena Blanco (10.2±0.0%). A very high severity cover of 76–100% was found affecting the secondary (3.3±1.4%) and primary branches (6.7 ±2.3%) of affected plants. A low cover of severity (≤25%) was observed in all regions of the thalli: apex (10.0±2.8%), secondary branch (2.5±1.4%), and primary branch (2.5±1.4%) (Fig. 5). Farms with the highest prevalence of EFA were recorded in Arena Blanco (11.2 ±0.4%), Guindacpan (4.5±1.7%), and Tigtabon (3.7 ±0.4%). All regions of the thalli, i.e., the apex (9.5 ±3.2%), secondary (16.7±4.1%), and primary (6.0±2.6%) branches, were affected; however, the severity cover was low (Fig. 6).

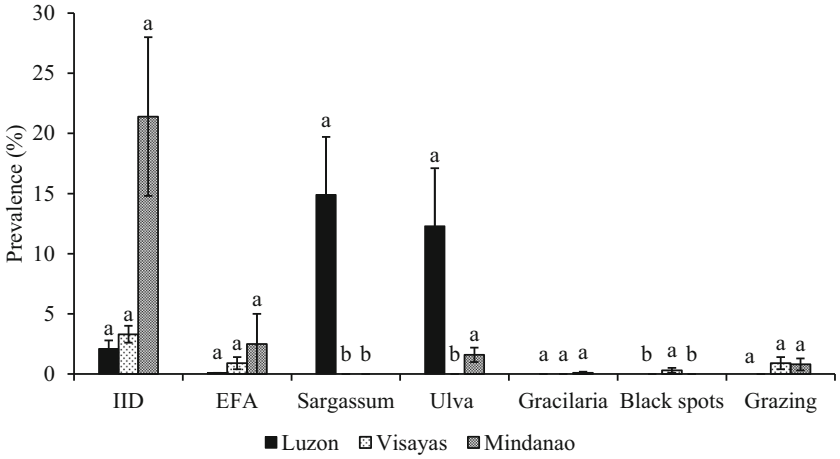


Fig. 3 Summary of the presence of "ice-ice" disease (IID), pests (macro-epiphytes and EFA), black spots, and grazing prevalence in 16 farms assessed: Luzon (n=5), Visayas (n=5), and Mindanao (n=6). Error bars signify ±SEM. Different letters denote significant difference at p<0.05

Environmental parameters

Environmental parameters varied between each farm (Table 4). Seaweed were exposed during low tide in shallow water farms, while for farms at deep water sites the depth ranged from 2.5 to 8.0 m during assessment. Irradiance ranged from 120 to 2670 μmol photons m^{-2} s^{-1}, with salinity in the range from 27 to 34 psu, temperature from 27.3 to 32.2°C, pH from 7.6 to 9.1, and dissolved oxygen from 5.3 to 9.2 ppm. The variation in readings of the different parameters will be influenced by the time of day and weather conditions at the time of sampling: for example, irradiance readings in Bohol were high because of intense heat and clear skies during the sampling period. For the nutrient analyses at the farms, phosphate levels ranged from not detected (n.d.) to 0.15 ppm, nitrite (0.04–0.07 ppm), nitrate (0.10–0.38 ppm), and ammonia (n.d. to 0.38 ppm). The highest volume of rainfall during the sampling month was recorded in Visayas farms (Bohol) with 223.6 mm, and the lowest monthly rainfall was observed in Samal, Davao del Norte (Mindanao) with 60.8 mm. The mean monthly wind speed ranged from 2–3 m s^{-1}. The near-surface air temperature (T_{mean}) ranged from 28.4 to 29.0, with the highest recorded at Samal farm.

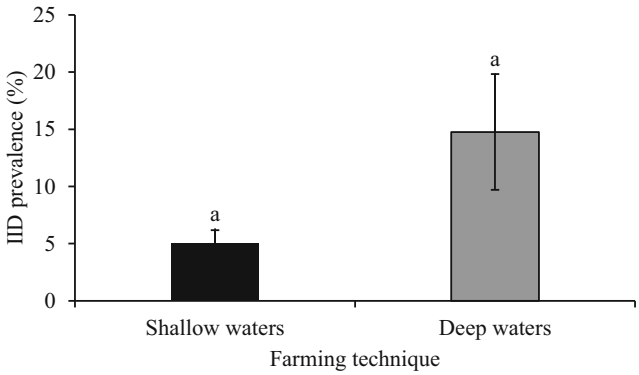


Fig. 4 "Ice-ice" disease (IID) % prevalence in relation to the farming techniques employed (n=16). Error bars signify ±SEM

Discussion

This preliminary survey is the first attempt to assess the prevalence of major yield-limiting pests, diseases and other abnormalities in eucheumatoid farming areas in the Philippines. Our results revealed differential prevalence of major diseases such as IID at farms across the country. Notably, IID is the most prevalent condition, affecting all farms assessed in this study, indicating the likely widespread occurrence of this at national level within the Philippines. This confirms previous reports of IID occurrence in the country, but it is the first to examine the prevalence of IID outside of specific provinces or regions (Uyenco et al. 1981; Largo et al. 1995; Hurtado et al. 2006a; Solis et al. 2010; Alibon et al. 2019). Although relatively low prevalence rates were recorded, it is of note that IID was observed in all the farms visited during the study, spanning an array of geographic locations, culture techniques employed, cultivation densities and cultivars used. The exact nature of the complex interactions between environmental and biotic factors leading to IID remains unknown; however, the occurrence of IID across a range of cultivation techniques and water depths suggests these relationships are not straightforward. All sites surveyed are within highly productive seaweed cultivation areas in Luzon, Visayas, and Mindanao and the extent to which the presence of IID at one site influences the occurrence at sites close by remains to be determined. A very high IID prevalence was observed in a deep water long line seaweed farm in Mindanao. Although the site was assessed only once, the low total monthly rainfall and wind speed recorded in the area in comparison to the other farms assessed might contribute to the exceptionally high IID prevalence. Low rainfall volume might result in high surface water temperature and increased salinity, while low wind speed might result in slow water movement. High prevalence of IID has previously been linked to high water temperature, salinity, and pH (Arasamuthu and Edward 2018; Msuya and Porter 2014; Alibon et al. 2019).

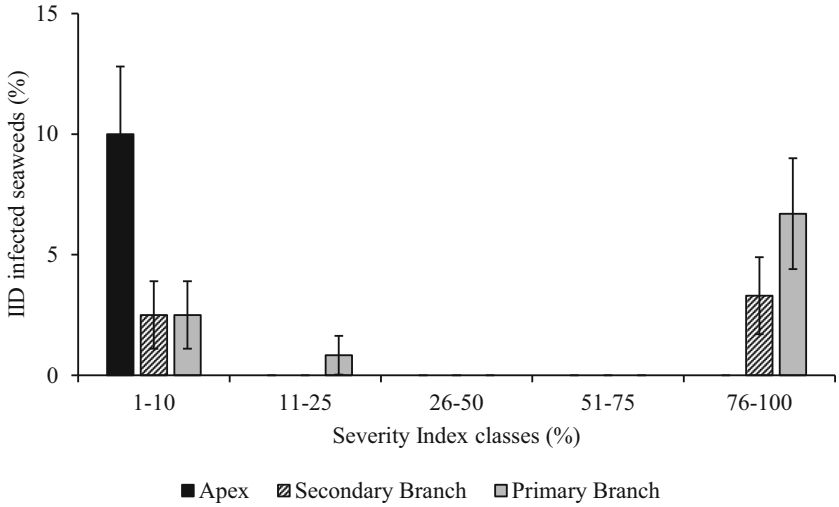


Fig. 5 "Ice-ice" disease (IID) severity cover (%) on the thallus regions of the infected seaweeds (±SEM; *n*=120)

Macro-epiphyte pests, including *Sargassum*, *Ulva*, and *Gracilaria*, are the common macroalgae found on the cultivation lines. The prevalence of *Sargassum* was highest at sites in Taytay (Luzon) in October 2019 (26–50%) where *Kappaphycus* is farmed in the lower intertidal area, in close proximity to *Sargassum* beds. In the Philippines, the annual vegetative growth peaks of *Sargassum* are observed during the months of October to December (Trono and Largo 2019), which coincides with high prevalence of pest fouling of cultivated eucheumatoids in this area. This high prevalence of *Sargassum* was not observed in other farming sites in Visayas and Mindanao which utilized the same farming types and were sampled at seasonally equivalent dates, suggesting that other factors, including slow water movement and anthropogenic eutrophication, might contribute to the proliferation of macro-epiphytes. The location of the farms in Taytay (Luzon) is within a bay which protects the farm sites from the strong currents and wind actions, preventing the macro-epiphytes from dispersing once they are already entangled in the cultivation lines. Also, human settlements and effluent outflow may contribute to eutrophication in the farming sites that could lead to the seasonal bloom of other macro-epiphytes, especially *Ulva* (Largo et al. 2004; Teichberg et al. 2010).

A limiting factor in recording EFA, black spots, and grazing in situ is the challenge of accessing seaweeds in deep water farms where the seaweeds remain below the water level at all times. However, the low prevalence rates recorded for EFA and grazing could be related to the seasonality in occurrence. For EFA, fluctuation in environmental conditions and other biotic factors may contribute to their low prevalence. The high volumes of rainwater during the wet season, which coincides with the date of sampling activities (July–November), may have contributed to a decrease in the salinity of waters around the farm and may account for the low prevalence of EFA. This observation is reflected in the study of Vairappan (2006) in Malaysia, where a lower prevalence of EFA was observed in September to November when seawater temperature dropped from 30 to 25°C and the salinity from 29 to 27 psu. The incidence of EFA was observed to be higher during the dry months (March–June), when the seawater temperature and salinity increases from 27 to 31°C and from 28 to

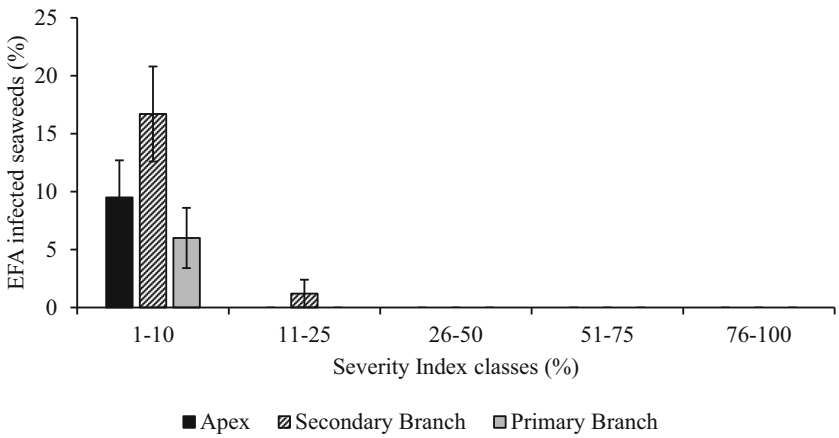


Fig. 6 Epiphytic filamentous algae (EFA) severity cover (%) on the thallus regions of the infected seaweeds (±SEM; *n*=84)

Table 4 Summary of environmental parameters of different farms assessed in major seaweed farming areas in the Philippines: Luzon (n=5), Visayas (n=5), and Mindanao (n=6). Monthly total rainfall data, mean wind speed, and mean near-surface temperature (T_{mean}) were collected from the nearest synoptic station of the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA)). *n.c.*, no data collected; *n.d.*, not detected; *e.c.*, exposed culture lines; ±SEM was not shown

Sampling sites	Date assessed	Time of sampling	Depth	Irradiance	Salinity	Temp.	pH	DO	Phosphate	Nitrite	Nitrate	Ammonia	Total rainfall (monthly)		Wind speed (monthly)	Near- surface air temp. (monthly)
			m	μmol photons m^{-2} s^{-1}	psu	°C		ppm	ppm	ppm	ppm	ppm	1 month before sampling mm	Sampling month mm	m s^{-1}	T_{mean} °C
Luzon																
Palawan																
Puerto Princesa	Oct 5, 2019	9:00 am	6.0	124	33	28.8	7.8	6.5	0.14	0.06	0.29	0.12	254.4	122.8	3	28.6
Balabac	Oct 6, 2019	10:30 am	e.c.	n.c.	n.c.	n.c.	8.6	n.c.	0.13	0.06	0.20	0.08	254.4	122.8	3	28.6
Quezon	Oct 7, 2019	10:00 am	2.5	120	34	32.2	7.8	7.2	0.15	0.05	0.37	0.06	254.4	122.8	3	28.6
Pamantolon, Taytay	Oct 8, 2019	9:00 am	n.c.	131	33	n.c.	7.7	n.c.	0.08	0.07	0.38	n.d.	300.2	190.6	2	28.4
Pularaquen, Taytay	Oct 9, 2019	3:00 pm	n.c.	n.c	n.c.	n.c.	n.c	n.c.	n.c.	n.c.	n.c.	n.c.	300.2	190.6	2	28.4
Visayas																
Bohol																
Danahon Reef	Oct 14, 2018	8:00 am	e.c.	2656	31	29.3	7.6	5.3	0.04	0.04	0.17	0.38	195.7	223.6	3	28.6
Mahanay	Oct 16, 2018	9:20 am	e.c.	1234	30	31.0	7.8	6.5	0.04	0.04	0.14	0.07	195.7	223.6	3	28.6
Guindacpan	Oct 16, 2018	9:00 am	n.c.	1278	30	31.0	7.8	6.5	0.04	0.04	0.14	0.07	195.7	223.6	3	28.6
Jao 1	Oct 17, 2018	10:00 am	e.c.	2670	28	31.6	7.8	6.1	0.05	0.04	0.26	0.05	195.7	223.6	3	28.6
Jao 2	Oct 17, 2018	10:00 am	e.c	2670	28	31.6	7.8	6.1	0.05	0.04	0.26	0.05	195.7	223.6	3	28.6
Mindanao																
Zamboanga City																
Tigtabon	Jul 24, 2019	11:40 am	4.7	n.c.	27	27.9	8.8	6.6	0.08	0.05	0.27	0.02	48.5	115.2	2	28.8
Layag-Layag	Jul 23, 2019	10:00 am	8.0	n.c.	32	27.3	8.9	6.8	n.d.	0.07	0.31	0.09	48.5	115.2	2	28.8
Arena Blanco	Jul 25, 2019	10:00 am	4.3	1685	30	27.3	9.1	6.2	0.08	0.07	0.33	0.07	48.5	115.2	2	28.8
Davao del Norte																
Samal	Nov 11, 2018	8:00 am	n.c.	n.c.	34	29.0	7.7	5.9	0.07	0.04	0.10	0.05	142.7	60.8	2	29.0
Tawi-Tawi																
Simunul	Nov 14, 2018	12:10 pm	e.c.	737	29	29.1	7.8	9.2	0.08	0.04	0.12	0.13	n.c	n.c.	n.c.	n.c.
Tundon	Nov 13, 2018	11:00 am	e.c.	2149	30	29.0	7.9	9.1	0.06	0.06	0.11	0.12	n.c.	n.c.	n.c.	n.c.

34 psu, respectively. The same result was observed in the study of Hurtado et al. (2006a), where outbreaks of EFA in the Philippines were noticed in farms having high temperature coupled with high irradiance and slow water movement. Although no direct correlation was observed between the environmental water parameters monitored and disease and pest prevalence in the present study, the variations in the data (e.g., higher, or lower levels of irradiance, temperature, and salinity) could potentially trigger the occurrence of EFA. This further supports the hypothesis that extreme fluctuations or abrupt, significant environmental changes are predisposing factors for the occurrence of pests and diseases and have a direct impact on the susceptibility of the cultured seaweeds to these problems. The association of pest and disease prevalence related to these time-point and seasonal variations in key environmental parameters warrants more extensive investigation than was possible in the current study.

The relatively high prevalence of IID observed in farms located in deeper waters than in shallow water farms, albeit not a significant result, was unexpected as the shallow water farms are more exposed to fluctuations in environmental conditions. Many studies of IID to date report on the occurrence of the condition in shallow water farms, and there is a lack of information on the occurrence and progression of the condition in deeper water farms, and no information available on the occurrence of IID outside cultivation settings. It may be that prolonged high or low rainfall volume, low wind speed and high near-surface temperature in deep water cultivation sites contribute to a stress response in seaweeds resulting in the development of disease signs. Changing weather patterns brought about by climate change may also influence seaweed disease occurrence in deep water farms. The results of this study may provide new insight that IID is no longer confined in shallow water farming areas but also occurs in deep water farms when unfavorable environmental conditions continue to persist over an extended period of time.

The presence of EFA and IID in the farms assessed suggests that these problems may occur regularly but have previously gone unreported. In this study, Mindanao reported the highest prevalence of EFA and IID compared with Luzon and Visayas regions. High EFA prevalence and high IID prevalence was also observed in Zanzibar (Tanzania) (Largo et al. 2020). Previous reports in the Philippines suggested that fluctuations in environmental conditions, including elevated seawater temperature, resulted in increased IID and EFA occurrences (Hurtado et al. 2006a; Alibon et al. 2019; Vince Cruz-Abeledo et al. 2019). Anecdotal reports of EFA and IID outbreaks on farms in the country have been noted, especially during prolonged extreme environmental conditions (e.g., El Niño and La Niña). Pang and Liu (2019) described that IID and EFA outbreaks in eucheumatoid farms in China are directly related to these extreme environmental anomalies together with the data obtained from El Niño Southern Oscillation (ENSO). This information provided insights on the effects of changing climate patterns to the occurrence of diseases and pests. Recently, Largo et al. (2017) discussed the direct and indirect impacts of global climate change as well as the effects of El Niño and La Niña in farmed eucheumatoids. These climatic events resulted in fluctuations in environmental parameters making the seaweeds more vulnerable to disease and pest outbreaks. The occurrence of typhoons and strong storm surges in farming areas could further negatively affect production. In Tanzania, the recorded extreme increase in surface water temperature correlates to a high incidence of IID and EFA, leading to die-offs of seaweeds mostly located in shallow water farms (Msuya and Porter 2014; Largo et al. 2020). Fluctuation in weather conditions brought about by climate change could further exacerbate factors associated with reduced health status in seaweeds, and this may prevent year-long continuous seaweed production, resulting in a decreased biomass production and reduced income for stakeholders.

IID was observed in different parts of the thallus, suggesting that any part of the seaweed is susceptible to IID. Changes observed in affected seaweeds belong to association type 1 (changes to thallus tissue). IID-affected seaweeds have previously been found to harbor purportedly pathogenic microbial complexes including gram-negative bacteria (*Vibrio* sp., *Cytophaga-Flavobacterium* complex, *Alteromonas*, *Pseudoalteromonas*, and *Aurantomonas*), and fungi (*Aspergillus ochraceus*, *A. terreus*, *Phoma* sp.) (Largo et al. 1995; Solis et al. 2010; Syafitri et al. 2017). *Vibrio* spp. have also been shown in vitro to induce more severe IID in healthy tissue in the presence of an environmental stressor, specifically low water salinity (Largo et al. 1995) and elevated water temperature (Azizi et al. 2018). These results may imply that environmentally stressed seaweeds are more susceptible to colonization by pathogenic bacteria, however no data exist relating to the full diversity of microbial taxa associated with eucheumatoids (in either healthy or diseased states), and the processes behind the development of IID signs remain unknown. Black spots were observed only in one farm, and the occurrence and effect of this abnormality in farmed seaweeds is as yet unknown. To the best of our knowledge, this is the first described incidence in the Philippines. Black spots warrant further studies to fully understand and establish whether their presence could be a potential problem for the seaweed industry in the future.

Macroalgae including *Sargassum*, *Ulva*, and *Gracilaria* belong to the association type 2, where the pest attaches loosely to the surface of the host, but no damage is observed (Leonardi et al. 2006; Ingle et al. 2018). However, entangling of these macro-epiphytes may result in compromised growth of farmed seaweeds though competition for nutrient uptake from the water and photosynthesis

may occur. EFAs are categorized under the association type 3 where the epiphytes penetrate the host seaweeds intercellularly, thereby injuring the cortical and medullary cells (Ask 1999; Vairappan 2006; Pang et al. 2011; Tsiresy et al. 2016). The EFA severity cover was observed to be low, affecting the whole thallus of the plant. There was no difference in the severity cover between thallus regions, but Yamamoto et al. (2012) reported that the soft tissues of the apex thallus are more susceptible to epiphyte colonization, suggesting that the initial entry of the epiphytes is in young tissues. When observed microscopically, EFA penetrated up to the cortical cells and created injury to the host plant (Vairappan 2006), thereby providing an entry point for pathogenic microbes, resulting in secondary infection. EFA outbreaks in seaweed farms may result in biomass loss and compromised carrageenan quality (Correa and McLachlan 1994; Hurtado et al. 2006; Vairappan et al. 2008; Ali et al. 2018, 2020).

Pang et al. (2015) observed the seasonal consumption of eucheumatoid crops by herbivorous fish, especially siganids (rabbitfish). Absence of tips on the soft tissues at the apex of the thallus is observed in the affected seaweeds (Ask 1999). Grazing of the tips of seaweed thalli was also observed in our study (association type 4), suggesting that juvenile herbivorous fish commonly consume seaweeds in intertidal areas where the seaweed farms are located. Grazing damage to seaweed thalli has been shown in some cases to significantly shift microbial community structure (Tan et al. 2020), which may reduce the health status of affected plants. Grazing also results in a loss of biomass and therefore a production decline in the farmed seaweeds. The protection of seaweed crops from grazing fish may be a beneficial management intervention to protect stock from damage which leads to incursion of environmental pathogens which further limit growth and yield.

Differences in the prevalence of diseases and pests between the eucheumatoid species assessed in this study were not captured. However, the previous study of Pang et al. (2015) identified *E. denticulatum* as being less susceptible to IID and EFA when compared to *K. alvarezii* and *K. striatus*. Also, in the same study, it was observed that a lower incidence of IID and EFA in farms with co-cultured conditions (i.e., *Kappaphycus* spp. with *E. denticulatum*). *E. denticulatum* has been found to release volatile halogenated compounds (VHCs) under stressful conditions, which might aid as a defense mechanism against potential harmful organisms (Collen et al. 1995; Mtolera et al. 1996). The VHCs produced by *E. denticulatum* might explain the high resistance of *E. denticulatum* to EFA and IID than *Kappaphycus* spp. (Pang et al. 2015). Multi-species methods of culturing seaweeds either in shallow and in deep water farms could potentially minimize the occurrence of disease and pest problems.

Conclusion and recommendations

This is the first study to compare the occurrence of important eucheumatoid diseases and pests across major seaweed farming areas in the Philippines. We also report the occurrence of black spots in farmed seaweeds for the first time. IID occurred in all regions surveyed, and across all cultivation methods and cultivars sampled. In contrast, macro-epiphytes and EFAs were not reported at all sites and when present were typically at moderate to low levels. However, each site was sampled only once, and therefore it remains to be seen how the prevalence of pests and diseases changes over growth cycles and seasons, and how this responds to fluctuations in environmental conditions including salinity, water temperature, and nutrient availability. It should be a priority for future research to address this lack of knowledge. Further studies should also aim to increase our understanding of the relationships between eucheumatoids and their associated microbial communities (including prokaryotes, protozoans, fungi, and viruses), the structure and function of which may play a role in the development of disease signs. A greater understanding of the factors influencing disease and pest occurrence will inform proactive management strategies, which are becoming increasingly important as disease and pest outbreaks become more frequent.

Biosecurity measures, such as quarantine procedures, should be imposed when translocating cultivars, particularly in major seaweed growing areas where the disease and pest transmission potential are high, in order to prevent the transfer of unchecked cultivars to other farming sites around the country. The inclusion and the strict implementation thereafter of biosecurity protocols in the Code of Good Aquaculture Practices for Seaweeds (GAqP-S) (Philippine National Standards/Bureau of Fisheries Standards (PNS/BAFS 208: 2017)) will be an important initial step in securing the future of the industry. Ongoing research and government initiatives coupled with seaweed stakeholders' support will help improve and sustain the industry, not only benefiting the aquaculture sector but also most importantly protecting the livelihood of thousands of farmers engaged in seaweed cultivation.

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